



POSTER PRESENTATION TOPICS

Junior Science Camp 2026 · Quantum, AI & Materials

None of these questions has a settled answer. Your poster must show **what is already known** about the physics, and then give **your team's own answer**, with your reasoning.

MATERIALS

- M1** Could we ever build a material that conducts electricity perfectly at room temperature?
- M2** When silicon finally runs out of room, what should replace it?
- M3** Could a sheet of carbon one atom thick change how we build everything?
- M4** Is there a limit to how much of the Sun's energy we can ever capture?
- M5** Could a battery ever hold as much energy as a tank of petrol?
- M6** Should we build computers out of light instead of electricity?
- M7** Could an object ever be made truly invisible?
- M8** Could a material be designed to heal itself, the way living tissue does?
- M9** Could we ever build a cable strong enough to reach into space?
- M10** Could a city run on the heat we currently throw away?

QUANTUM

- Q1** Classical physics worked so well — so why did it have to be replaced?

- Q2** Does an electron decide what it is only at the moment we look at it?
- Q3** Is the uncertainty principle a limit on our instruments, or on reality itself?
- Q4** Could a particle pass through a wall that you could not?
- Q5** Is randomness built into the universe, or do we simply not know enough yet?
- Q6** Could a quantum computer solve a problem that no ordinary computer ever could?
- Q7** Can a message ever be sent in a way that is impossible to intercept?
- Q8** Are quantum effects at work inside living things?

ARTIFICIAL INTELLIGENCE & COMPUTATION

- A1** Could a machine ever be conscious, and how would we know?
- A2** Could an AI discover a law of physics that humans have missed?
- A3** Will we ever be able to predict the weather a month in advance?
- A4** Should we trust a scientific result that no human can explain?
- A5** Could a computer invent a material that no human would have thought of?
- A6** Is there a physical limit to how much a machine can think?